The Federal University of Technology, Akure.
School of Engineering and Engineering Technology
Department of Electrical and Electronics Engineering
First Semester Examinations 2023/2024 Session
EEE 401: Introduction to Control Systems

Date: 18-05-2024

Instructions: ANSWER **FOUR** Questions in all.

Time Allowed: $2\frac{1}{2}$ HOURS

Unit: 3.0

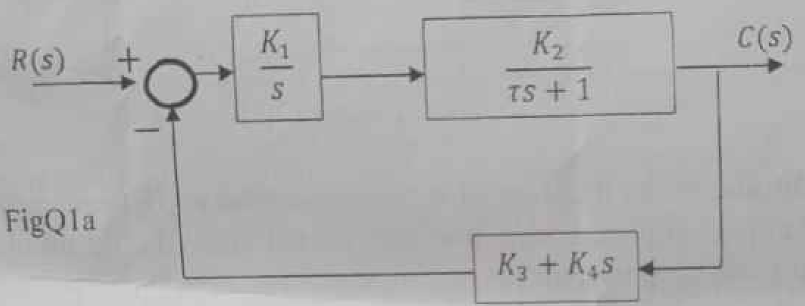
Question 1

a. Consider a control system that has a position and rate feedback as shown in FigQ1.

(i) Determine, the closed loop transfer function $\frac{C(s)}{R(s)}$. **(6 Marks)**

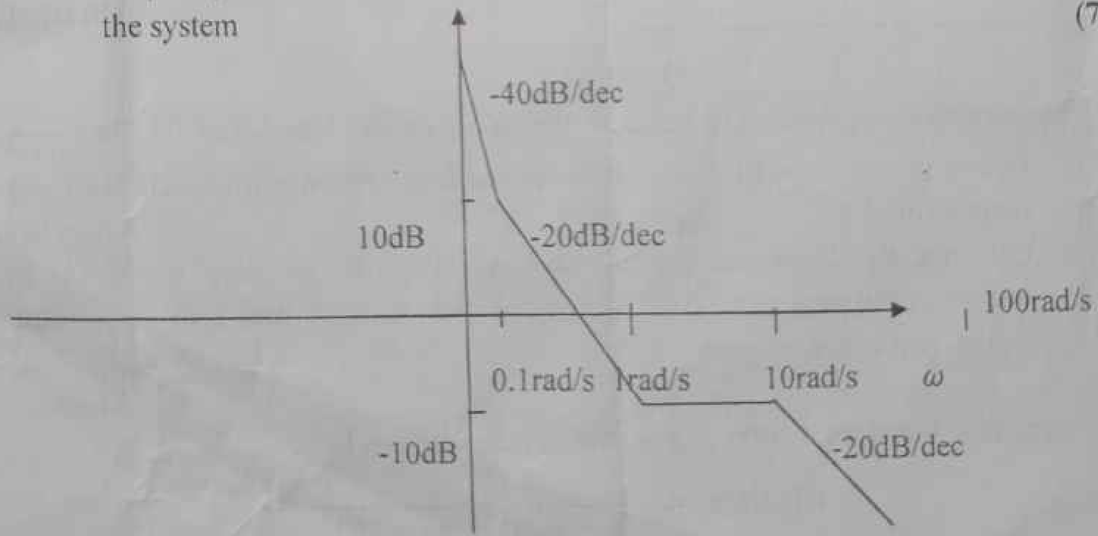
(ii) Determine the sensitivity of the closed loop transfer function to k_2 as a function of s. **(6 Marks)**

(iii) What is the sensitivity of the system in Question 1a(ii) at DC. **(6 Marks)**



FigQ1a

b. An engineer was called in to consult on a control system of a piece of equipment in the field. No one can find the design report or test results from the original design of the control system. The engineer therefore decided to take a frequency response of the control system by opening up the outer feedback loop. The resulting asymptotic gain-frequency characteristics is as obtained in FigQ1b. Obtain the loop transfer function of the system **(7 Marks)**



FigQ1b

Question 4

a. Given a unity feedback system of

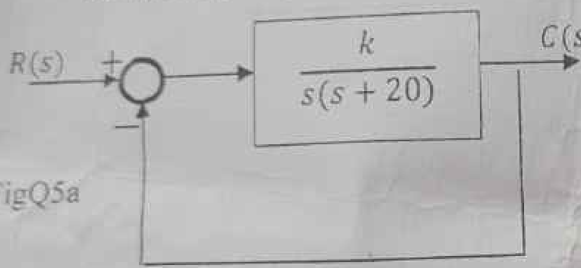
$$G(s) = \frac{10(s+10)}{s(s+2)(s+5)}$$

Give a bode plot of the system on a semi log graph with all the calculations clearly shown. Determine the **(a)** Phase Margin (PM), and **(b)** Gain Margin (GM) from the graph. **(13 Marks)**

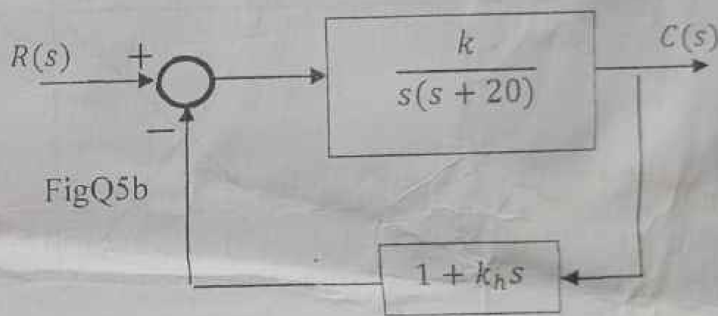
b. Proof that the Bandwidth of a closed loop control system given as $T(s) = \frac{\omega_n{}^2}{s^2+2\rho\omega_n s+\omega_n{}^2}$ is $\omega_n\sqrt{1-2\rho^2 \pm \sqrt{2-4\rho^2+4\rho^4}}$ **(12 Marks)**

Question 5

a. Determine the value of k when the damping ratio of a system is 0.158 and the undamped natural frequency is 3.16 rad/sec as shown in FigQ5a



FigQ5a



FigQ5b

In order to improve the relative stability of the system, a tachometer feedback was introduced as shown in FigQ5b. Determine the value of k_h so that the damping ratio of the system becomes 0.5 **(13 Marks)**

b. The control unit of a missile launcher is represented by an open loop transfer function $G(s) = \frac{10}{s(s+1)(s+4)}$ as a control consultant give a polar plot representation of the system using an appropriate graph. **(12 Marks)**

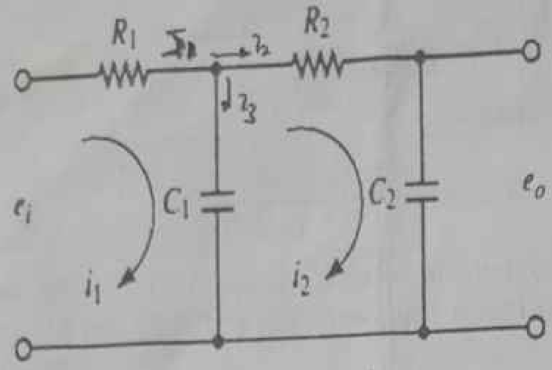
Question 6

a. Consider the electrical circuit shown in FigQ6.

(i) Obtain the equations for the circuit; **(5 Marks)**

(ii) Draw a block diagram from the equations obtained in (i); and **(3 Marks)**

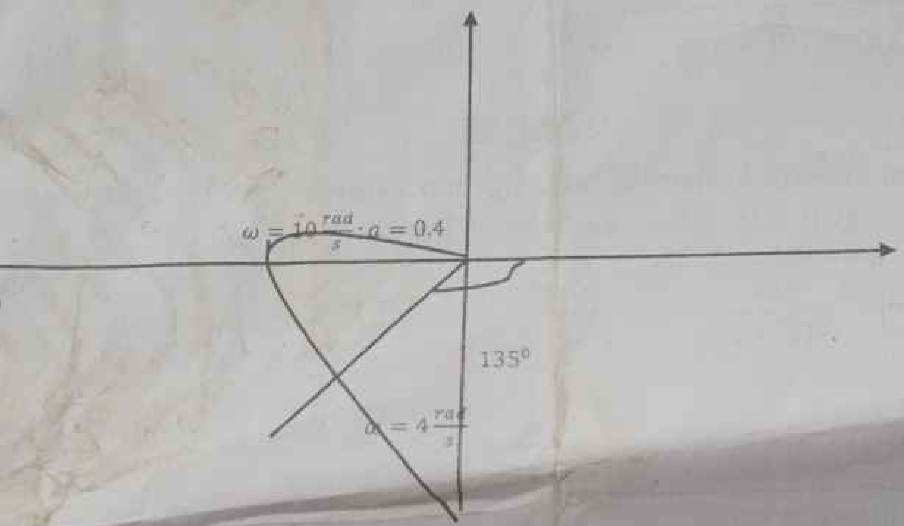
(iii) Obtain the transfer function G(s) from the block diagram drawn in (ii). **(4 Marks)**



FigQ6a

b. A Nyquist diagram or Polar plot for a control system is shown in FigQ6b

FigQ6b



The control system is known to have an open loop transfer function whose form is given by the following

$$G(s)H(s) = \frac{k_1}{s^3 + k_2 s^2 + k_3 s}$$

Determine the values of k_1, k_2, k_3 **(13 Marks)**

THE FEDERAL UNIVERSITY OF TECHNOLOGY AKURE

School of Engineering and Engineering Technology

Computer Engineering Department

2023/2024 First Semester Examination

Degree: B.Eng
Course Title: Assembly Language Programming
Course Code: CPE 405
Time: $2\frac{1}{2}$ Hour
Instruction: (i) Attempt question one (1) and any other four
(ii) The of use any form of **calculator, phone and smart/digital wrist watch** is prohibited

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1. (a) Explain what a linker is and; show if and why it is needed in assembly language programming

(3 marks)

(b) Highlight how your skill in assembly language programming will favour your knowledge of operating system (3 marks)

(c) Write a program to add two numbers in C and Assembly languages (8 marks)

(d) You wrote an assembly language program for a project in which Intel 8051 was to be used. After writing your program you realize you actually have Motorola 68000, ~~Intel 8085~~ & TMS 19900.

i. Which of these will you choose as alternative to implement your program?
ii. Give convincing reasons for your answer in (i)

(6 marks)

2. (a) i. What is addressing mode?

ii. List any three data transfer instruction with comments

(5 marks)

(b) "Assembly language has a one to many relationship with high level language whereas it has a one to one relationship with machine language", Elucidate. (5 marks)

3. (a) Comment succinctly on the following assembly language statements: (6 marks)

ADD A, byte
INC A
INC byte
INC DPTR

DEC A
DEC byte
MUL A, eax
CALL write Int

(b) (i) What is abstraction in programming?.

(ii) Use abstraction to relate low level language to high level language

(4 marks)

4. (a) Explain the functions of the following:

i. Stack pointer
ii. Program Counter
iii. Accumulator

(6 marks)

(b) Write an assembly language program to transfer data from one general purpose register staring at address 20H to another general purpose register starting at address 30H.

(4 marks)

5. (a) Specify the addressing mode of the following instructions: (4 marks)

i. MOV A, @ R1
ii. MOV R2, #60H
iii. MOV @ R4, A
iv. MOV R0, 6FH

(b) Compare Compiler and Interpreter, thus aptly explain how they affect debugging (5 marks)

6. (a) Discuss the assemble, link and execute cycle

(7 marks)

(b) List the disadvantages of machine language

(3 marks)

7. (a) Discuss instruction set architecture with examples

(3 marks)

(b) Write program to display a string ***"It's a good day"*** in C and assembly language (7 marks)

THE FEDERAL UNIVERSITY OF TECHNOLOGY AKURE

School of Engineering and Engineering Technology

Computer Engineering Department

2023/2024 First Semester Examination

Degree: B.Eng
Course Title: Microprocessors and Interfacing
Course Code: CPE 403
Time: $2\frac{1}{2}$ Hour
Instruction: (i) Attempt question one (1) and any other four
(ii) The of use any form of **calculator, phone and smart/digital wrist watch** is prohibited

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1. (a) You are given an embedded system project to develop a system to perform a ask:

   i. Name and explain the type of processor you will use

   ii. What are the reason(s) for choosing the processor

   iii. What differentiate it from other processors

   (6 marks)

   (b) Discuss the evolution of Microprocessors (6 marks)

   (c) Discuss exhaustively, the computer memory hierarchy (6 marks)

   (d) Describe the reduced instruction set computer RISC (2 marks)

2. (a) i. Give a differential explanation of the following memory devices:

   Cache, RAM & Register.

   ii. Succinctly discuss the meaning and functions of Program Counter (6 marks)

   (b) With the aid of a simple but relevant diagram, describe the Program Status Word

   (4 marks)

3. (a) Keenly explain different data transfer modes in computer (6 marks)

   (b) State the household applications of microprocessor (4 marks)

4. (a) In a tabular form, explain the difference between reduced instructions set computer RISC and complex instructions set computer CISC. (5 marks)

   (b) Explain the bus architecture of Intel 8086 (5 marks)

5. (a) Why do you need time sharing in multitasking (2 marks)

(b) List the pros and cons of 8086 architecture (8 marks)

6. (a) What are the concerns of a real-time system (6 marks)

(b) Discuss interrupt, hence scheduling in interrupt system (4 marks)

7. (a) With the aid of a well labeled diagram, explain 8085 microprocessor architecture. (8 marks)

(b) Discuss compatibility issue that may arise when interfacing I/O devices with microprocessor (2 marks)

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SECTION B

2. Discuss the following addressing modes with a suitable example:
   - (i) Immediate Mode
   - (ii) Index Mode
   - (iii) Indirect Mode
   - (iv) Absolute (Direct) Mode
   - (v) Register Mode

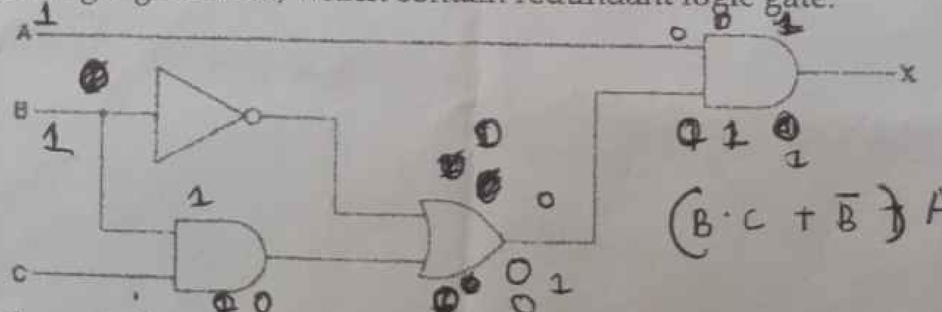
(20 Marks)

3. (a) Describe 3 steps in 'Fetch Operation' of instruction cycle. (5 Mark)
   (b) Discuss "Cache Memory" (3 Mark)
   (c) Briefly explain "Cache Controllers" and its features (4 Mark)
   (d) Discuss the following:
      (i) Real (Physical Memory) and its technique (4 Mark)
      (ii) Semiconductor Memory and its properties (4 Mark)

4. (a) What are the compositions of an "Intel Instruction ". (4 Mark)
   (b) Briefly explain the following "Intel 8085 Instructions". (6 Mark)
      (i) **ADD** ***r*** ***with example.***
      (ii) **RAL**
   (c) Mention 4 alternative ways of classification of "Instructions". (4 Mark)
   (d) Explain classification of Processors/Computers (6 Mark)

5. (a) Briefly compare features of INTEL 8085 and INTEL 8086 (5 Mark)
   (b) Discuss "Combinational and Sequential circuits" with example (6 Mark)
   (c) Present Half-Adder (Arithmetic Circuit) using Exclusive-OR. Hence, support your presentation with a truth table. (5 Mark)
   (d) With aid of diagram, compare RISC and CISC architecture (4 Mark)

6. Consider the following logic circuit, which contain redundant logic gate.



$(B \cdot C + \bar{B})A = X$

ABC +

   (a) Write the Boolean algebraic expressing of the above circuit (3 Mark)
   (b) Develop truth table for the logic circuit (8 Mark)
   (c) Develop K-map for the truth table in (b) to produce an optimal sum-of-products and realize its equivalent logic gate (9 Mark)

BEST OF LUCK

The Federal University of Technology, Akure.
School of Engineering and Engineering Technology
Department of Electrical and Electronics Engineering
First Semester Examinations 2023/2024 Session
EEE 405: Communication Principle **Date: 17-05-2024**

Instructions: ANSWER **FOUR** Questions in all.

Time Allowed: $2\frac{1}{2}$ HOURS

Unit: 3.0

Question 1

(13 Marks)

a. Derive the Frequency Modulation (FM) expression

b. Consider a receiver cascade. The Low Noise Amplifier (LNA) has gain $(G_1) = 20$ dB and Noise Figure $(NF_1) = 2$ dB. The mixer has a communication gain $(G_2) = 100$ dB and Noise Figure $(NF_2) = 10$ dB. The Video Graphic Array (VGA) has gain $(G_3) = 50$ dB and Noise Figure $(NF_3) = 15$ dB. Calculate the overall Noise Figure of the receiver **(12 Marks)**

Question 2

a. Describe briefly **(i)** Frequency Division Multiplexing **(ii)** Time Division Multiplexing and **(iii)** Frequency Hopping **(10 Marks)**

b. The antenna current of an AM transmitter is 8 A when only the carrier is sent, but increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage (%) modulation. Also, determine the antenna current when the depth of modulation changes to 0.8. **(15 Marks)**

Question 3

a. Explain with relevant equations Vestigial Side-band Amplitude Modulation (AM) communication system **(10Marks)**

b. (i) Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100 percent and (b) 50 percent **(9Marks)**

(ii) Suppose a Wi-Fi router operates in a noisy environment with bandwidth of 20MHz. The signal power transmitted by the router is 100mW and the noise power in the channel is 1mW. Calculate the Signal to Noise Ratio (SNR) and Channel Capacity **(6 Marks)**

Question 4

a. Explain the three functional components of a communication system in details using relevant block diagrams **(12 Marks)**

b. Derive the frequency components in a double side band suppressed carrier (DSB-SC) showing its signal strength obtained from the modulation of a carrier signal $10\cos(2\pi 10t)$ and message signal $\cos(2\pi t)$. **(13 Marks)**

Question 5

a. (i) Define the terms with relevant equations:

- Noise figure
- Noise factor **(10 Marks)**

(ii) Describe the working principle of a Phase Locked Loop (PLL) illustrating all the constituent block with their respective equations **(5 Marks)**

b. A 400W carrier is modulated to a depth of 75%. Calculate the total power in the modulated wave. **(10 Marks)**

Question 6

a. A semiconductor laser couples 5 mW into an optical fibre link. The optical signal travels through a group of components such as cable, connectors, splitters which gains are specified in Figure Q6a. compute the power input into the optical receiver. **(13 Marks)**

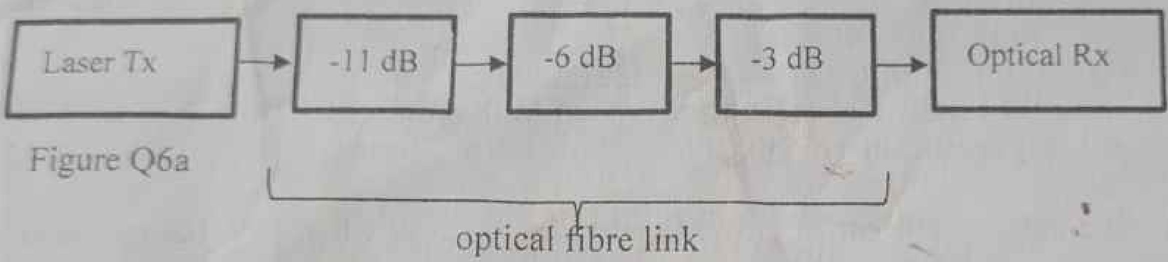


Figure Q6a

b. Explain a demodulation technique for conventional AM signal **(12 Marks)**

FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF COMPUTER ENGINEERING

First (1st) Semester Examination | **2023/2024 Academic Session**
Course Title: Software Development Techniques **(3 Units)** | **Course Code:** CPE 407/ICT 303
Instruction: Attempt ANY five (5) questions. | **Time:** 2 Hours 15 Minutes

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Question 1

(a) A computer's Central Processing Unit (CPU) cannot directly execute instructions that are written in a language like a spoken language. Explain why this is the case. Give TWO (2) **types** of software tools that can create machine-code instructions. **4 Marks**

(b) Identify **and** briefly explain THREE (3) qualities that a good algorithm should have. **6 Marks**

(c) Distinguish between an algorithm, pseudocode, and a flowchart. **6 Marks**

(d) Write an algorithm and draw a flowchart to convert temperature from Celsius to Fahrenheit. **4 Marks**

Question 2

(a) Describe the Rapid Application Development (RAD) model. **3 Marks**

(b) Differentiate between V-model and Big Bang model. **4 Marks**

(c) With the aid of a diagram, explain the FOUR (4) phases in spiral model. **6 Marks**

(d) Which life cycle model would you follow for developing software for each of the following applications? Justify your selection of model with appropriate reasons.

i. A financial application ii. A new text editor **4 Marks**

(e) How can software crisis be eliminated in the software development process? **3 Marks**

Question 3

(a) What is the difference between Verification and Validation? **3 Marks**

(b) How are requirements validated? **2 Marks**

(c) What is the major distinction between user requirement and system requirement? **2 Marks**

(d) As a software developer hired by a client to build an e-commerce website, list 5 things that are true about the requirements you get from the client. **5 Marks**

(e) What is an SRS document? State and explain 5 characteristics of a good SRS document. **8 Marks**

Question 4

(a) What is Unified Modeling Language (UML)? **2 Marks**

(b) List five main symbols shown on a Class Diagram and explain any two of them in detail. **8 Marks**

(c) What is a use-case diagram? **2 Marks**

(d) The following are the activities of a borrower in a Library System. Construct a use case diagram from these activities.

All the best

Borrow Books
Do Research
Reading Newspapers and Periodicals
Get Loan
Check Library Card
Return Books. **8 Marks**

Question 5

(a) Distinguish between a program and a software product. **3 Marks**
(b) What are the two types of software products? **2 Marks**
(c) What are the advantages and disadvantages of iterative software development model? **4 Marks**
(d) Why does software fail after it has passed from acceptance testing? **3 Marks**
(e) What are the testing principles the software engineer must apply while performing the software testing? **4 Marks**
(f) What are the two levels of testing? List the various testing activities. **4 Marks**

Question 6

(a) **For the scenario described below, which life cycle model would you choose? Give the reason why you would choose this model.**
You are interacting with the MIS department of a very large oil company with multiple departments. They have a complex regency system. Migrating the data from this legacy system is not an easy task and would take a considerable time. The oil company is very particular about processes, acceptance criteria and legal contracts. **5 Marks**

(b) **What are the principles of agile methods?** **5 Marks**
(c) What are the eight (8) levels of software that separates users from hardware? **4 Marks**
(d) Explain the Waterfall model and outline its various phases. **6 Marks**

Question 7

(a) Outline briefly what the following should do in a Scrum team.
i. Product Manager/Owner ii. Software development team iii. Scrum Master **6 Marks**
(b) List and explain the 3 types of software requirements. **7 Marks**
(c) Explain the benefits of using Agile methods like Scrum. **4 Marks**
(d) List three (3) core artifacts in any Scrum framework. **3 Marks**

 All the best

THE FEDERAL UNIVERSITY OF TECHNOLOGY AKURE

DEPARTMENT OF COMPUTER ENGINEERING

Title/code: Microprocessor/Assembly lang. Lab Test / CPE413/405

Time: 30 min

1. Write down the sequence of instruction. If the program begins with location 0400H and end at 0402 with HEX code B8 2F, and E6 00 respectively. find the data of the accumulator at the end of program

2. In 8086 microprocessor kit keyboard layouts, write the full meaning of the following abbreviations and its function:
   i. IP ii. DEL iii. REG vi. REP

3. Explain the following assembly language instruction:
   i. MOV C, B ii. CLC iii. JNC 2010 iv. INC B v. DEC C
4. Write an assembly language program to add two length numbers

5. Write a program with a given hexadecimal 3F and 1F add the two numbers 8bit to the accumulator register. Find the output of the program.